

Lab Report 9: Introduction to Photonic Quantum Computing

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1 Introduction

This lab gives us a basis in Boson Sampling using quantum photonic circuits. These circuits are simulated with Strawberry Fields. Boson sampling is a limited quantum computing task where indistinguishable photons are passed through a linear optical network consisting of beamsplitters and phase shifters. By analyzing interference patterns and detection probabilities at the output, we simulate quantum states and display the computational results associated with these interferences. The objective of this lab is to simulate an 8 mode circuit with polarized photons, analyze correlations, and use results to explain quantum interference.

2 Procedure and Results

Task 1 - HWP and Beam Splitter

We examine how a HWP alters polarization states using the H-V basis. In 1a the input is a horizontally polarized photon and in figure 1b the input is vertically polarized. The HWP aligned at 22.5 degrees is oriented to rotate the polarization axis by 45 degrees, producing diagonal polarization in both cases.

In both cases:

$$|D\rangle = \frac{1}{\sqrt{2}}(|H\rangle + |V\rangle)$$

In both cases, the HWP transforms the linear polarization state into an equal superposition of horizontal and vertical states. We examine the PBS. The PBS separates light based on polarization. Horizontally polarized light is

transmitted while vertically polarized light is reflected.

In general the input state for the PBS is:

$$|\psi\rangle = |1_H 1_V, 0_H 0_V\rangle$$

The PBS acts to split and redirect the two orthogonal polarizations into separate spatial modes.

Task 2 -PBS operator

In this task, we analyze how a PBS acts on a general two photon input state, specifically:

$$|\psi\rangle = |1_H 1_V, 0_H 0_V\rangle$$

This describes two photons, one horizontally polarized and the other vertically. Both are in mode 1. With the vacuum being in mode 0. The PBS routes photons based on their polarization. The horizontally polarized photons continue in their same spatial mode while the vertically polarized photons go into another spatial mode.

$$PBS|\psi\rangle = |1_H, 0_V\rangle \otimes |0_H, 1_V\rangle$$

$$|\psi'\rangle = |1_H 0_V, 0_H 1_V\rangle$$

Task 3- Gate Implementation

We determine the gate level implementation of the HWP and the PBS in the entanglement diagram provided in the lab report. The HWP acts as a transformation in the polarization basis. It maps the basis state to:

$$|H\rangle \rightarrow \frac{1}{\sqrt{2}}(|H\rangle + |V\rangle)$$

$$|V\rangle \rightarrow \frac{1}{\sqrt{2}}(|H\rangle - |V\rangle)$$

This operation can be implemented in the program with the `BSgate()` followed by phase shifts. In the Strawberry Fields simulations it is programmed as a mode transformation between two polarization modes.

This action can be modeled by a mode swap for vertical polarizations. This is represented by `BSgate( $\pi/2, 0$ )` or the swap matrix:

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

Task 4: Draw the Gates

Task 5: 8-mode program

Here we simulate an entanglement generation involving horizontally polarized photons using Strawberry Fields. The experiment simplifies the Bell state generation circuit. We start with defining an initial quantum state. One horizontally polarized photon in each of the four spatial paths. Since each path supports two polarization modes we define it as a 8 mode system. The quantum state is defined by a tensor shape of $[5]^8$. This correlated to the cutoff dimension of 5 photons per mode.

The initial state corresponds with:

$$|1_H\rangle_{A1} \otimes |1_H\rangle_{A2} \otimes |1_H\rangle_{B2} \otimes |1_H\rangle_{B1}$$

First each polarization pair experiences interference using a `BSgate`. Then the interferometer gates are applied across vertical modes to implement PBS swaps. Further `BSgate` and `BSgate` inverse operations recombine the modes. The complete network matches the Bell state circuit's functionality.

We found the norm of the ket to be equal to 1.0.

```
cutoff_dim = 5 # maximum number of photons + 1
paths = 4 # number of spatial paths
modes = 2 * paths # each path has two modes for polarization
initial_state = np.zeros([cutoff_dim] * modes, dtype=complex) # initialize the state tensor with zeros, with the shape defined by cutoff_dim and modes

# the ket below corresponds to a single horizontal photon in each of the modes
initial_state[1, 0, 1, 0, 1, 0, 1, 0] = 1 # set the initial state to have one horizontal photon in each pair of modes

# Permutation matrix for Swap Gate
X = np.array([[0, 1], [1, 0]]) # define a permutation matrix that swaps the two modes

# Here is the main program: create the input state and then send it through a network
prog = sf.Program(8)
with prog.context as q:
    Ket(initial_state) | q # apply the initial state to the quantum register
    for i in range(paths): # iterate over each path
        BSgate() | (q[2 * i], q[2 * i + 1]) # apply a BS gate to each pair of modes (within each path)
    Interferometer(X) | (q[1], q[3]) # apply an interferometer operation (permutation matrix) to modes 1 and 3
    Interferometer(X) | (q[5], q[7]) # " to modes 5 and 7
    BSgate() | (q[2], q[3]) # apply a BS gate to modes 2 and 3
    BSgate() | (q[4], q[5]) # " to modes 4 and 5
    Interferometer(X) | (q[3], q[5]) # apply an interferometer operation to modes 3 and 5
    BSgate().H | (q[2], q[3]) # apply the Hermitian conjugate (inverse) of the BS gate to modes 2 and 3
    BSgate().H | (q[4], q[5]) # " to modes 4 and 5
```

Task 6 - Probability of Given Event

To calculate the probability of a postselection event (in our case detecting one horizontally polarized photon in both path 2 and 3), we spit the ket tensor through our program to get the amplitudes to calculate probability.

This substate shows the event where detectors corresponding to modes 2 and 3 detect horizontally polarized photons. Taking the square amplitude of this even provides the probability of this specific postselection. The probability is sliced by computing the norm squared of the subket after isolating for mode 2 and 3 data.

Task 7- Success Probabilities Postselection

We expand this analysis to include all 6 possible coincidence patterns across the four detector modes. In order to do this we transpose the ket tensor to reorder the nodes so the indices correlate to detector outputs.

Then we define the six distinct postselection patterns where two horizontally polarized photons are detected across four modes.

For each pattern we extract the subket for data and find the probability of each event. The outputting result shows that each postselection pattern occurs with equal probability.

Results and Code for Task 6/7

```
The indices of the nonzero components for the six different postselections are
[[[0 0 1 1]
 [1 1 0 0]]

 [[0 1 0 1]
 [1 0 1 0]]

 [[0 1 1 0]
 [1 0 0 1]]

 [[0 1 1 0]
 [1 0 0 1]]

 [[0 1 0 1]
 [1 0 1 0]]

 [[0 0 1 1]
 [1 1 0 0]]]
The success probabilities for each pattern are the same
[0.03125 0.03125 0.03125 0.03125 0.03125 0.03125]
```

```
# transpose the ket vector for convenient
access based on the desired patterns
ket_t = ket.transpose(2, 3, 4, 5, 0, 1, 6, 7)
# arrange tensor index such that the 4
detector modes are in the first 4 slots of
the array

# define the postselection patterns,
specifying the presence (1) or absence (0) of
photons in specific modes
patterns = [
    [1, 1, 0, 0],
    [1, 0, 1, 0],
    [1, 0, 0, 1],
    [0, 1, 1, 0],
    [0, 1, 0, 1],
    [0, 0, 1, 1],
] # these are the 6 2-fold coincidence
patterns that are possible

# this returns the post-selected state of the
remaining two photons conditioned on a
particular detected pattern
sub_kets = [np.round(ket_t[tuple(ind)], 15)
for ind in patterns]

# calculate the success probabilities for
each post-selection pattern
# computed by taking the norm of each sub-ket
and squaring it
ps = np.array(list(map(np.linalg.norm,
sub_kets))) ** 2

# find the indices of the non-zero components
for each post-selected sub-ket
indices =
np.array([np.array(np.nonzero(sub_ket)).T for
sub_ket in sub_kets])

# print the indices of the non-zero
components for the six different
post-selections
print(
    "The indices of the nonzero components for
the six different postselections are \n",
    indices,
)
```

3 Discussion and Conclusion

This lab dived into how polarization can encode quantum information and how photonic elements such as HWPs and PBS can be modeled in Xanadu's Strawberry Field. Through the tasks, we started with basic polarization transformations and finished with the construction of an entangled photonic Bell state through postselection.

Task 1-3: Modeling Polarization Transformations

We started the lab by looking into the action of the HWP on horizontally and vertically polarized input photons. In both cases, we saw the polarization state was transformed into a 45 degree linear combination of horizontal and vertical states, producing a diagonal polarization state.

$$|D\rangle = \frac{1}{\sqrt{2}}(|H\rangle + |V\rangle)$$

The results from this part set up the rest of the lab by showing us how polarization transforms can represent quantum gates in the Strawberry Field software. We modeled the PBS functionality using a swap matrix acting on vertically polarized photons in the spatial field. The gate implementation of the following optical devices helped us create entangled Bell states.

Task 4-5: Circuit Simulation

The gate behavior was introduced. We implemented an entanglement generation circuit using beam splitters, interferometers, and horizontally polarized input photons. The input state for the simulation was:

$$|1_H\rangle_{A1} \otimes |1_H\rangle_{A2} \otimes |1_H\rangle_{B2} \otimes |1_H\rangle_{B1}$$

The state was first sent through a sequence of gates that act like the entanglement generation setup shown in the lab report. At the end of the simulation, the norm of the resulting ket vector was calculated for. We found that it was equal

to 1.0. This proved that the state was properly normalized for the chosen cutoff dimension. This finding was important because the normalized ket shows accurate probability predictions for the next tasks.

Task 6-7: Postselection Analysis

The postselection analysis showed us one of the most important insights in the lab. We first calculated the probability of detecting horizontally polarized photons in different paired paths. Even with different detector configurations, the probability of each of the six patterns was identical with a probability of 1/32.

The symmetry across all patterns suggests the output was highly symmetric. This proves that entangled Bell states generated by the circuit have no preference for detector modes. Every polarization path combination had an equal probability of capturing the entangled photons. We can attribute the symmetry to two points. First the uniform input modes had the same polarization states. Also, the beam splitters have a balanced nature which gives the photons symmetrical probabilities of reaching any output mode.

These findings confirm theoretical expectations about entangled photon behavior and validate the accuracy of our simulation. The postselection probabilities being symmetrical shows that our system preserved entanglement over various modes.